

Sri Lanka Institute of Information Technology

BSc. (Hons) Degree in Information Technology Specialized in Cyber Security

Department of Computer Systems Engineering

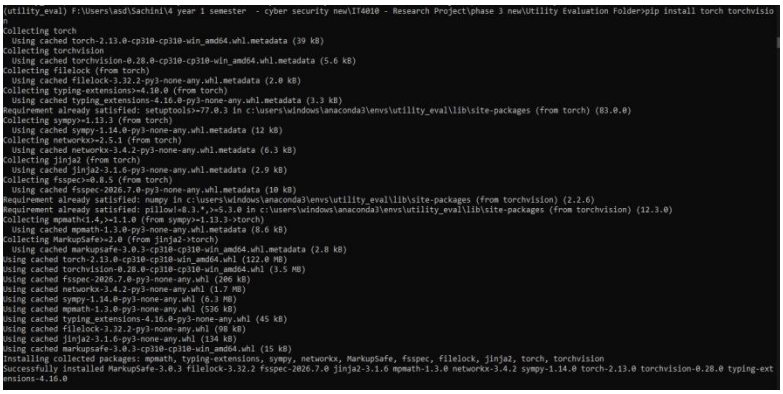
IT4010 – Research Project (Comprehensive Design and Analysis Project)

R26 – CS – 010

Privacy - Enhanced Synthetic Data Generation for Autonomous Vehicle Training

Research Project Periodic Log Entry Form

- **Project Id: - R26 – CS – 010**
- **Student ID: - IT22110220**
- **Name: - Apeksha M.K.S.R**

Date	Research Project Work
2026/08/01	• I set up the environment. • I installed **pillow 12.3.0** version and **numpy**.  • I installed **torch torchvision** and verify it.  • And also, I installed **ultralytics, pycryptodome, pillow, numpy, pandas, matploitlib, scipy**

```
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>pip install ultralytics pill
ow pandas matplotlib
Collecting ultralytics
  Using cached ultralytics-8.4.115-py3-none-any.whl.metadata (45 kB)
Requirement already satisfied: pillow in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (12.3.0)
Collecting pandas
  Using cached pandas-2.3.3-cp310-cp310-win_amd64.whl.metadata (19 kB)
Collecting matplotlib
  Using cached matplotlib-3.10.9-cp310-cp310-win_amd64.whl.metadata (52 kB)
Requirement already satisfied: filelock>=3.16.1 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (3.32.2)
Requirement already satisfied: numpy>=1.23.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (2.2.6)
Collecting opencv-python!=4.13.0.90,>=4.6.0 (from ultralytics)
  Using cached opencv_python-5.0.0.93-cp37-abi3-win_amd64.whl.metadata (20 kB)
Collecting pyyaml>=5.3.1 (from ultralytics)
  Using cached pyyaml-6.0.3-cp310-cp310-win_amd64.whl.metadata (2.4 kB)
Collecting requests>=2.23.0 (from ultralytics)
  Using cached requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Requirement already satisfied: torch>=1.8.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (2.13.0)
Requirement already satisfied: torchvision>=0.9.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (0.28.0)
Collecting psutil>=5.8.0 (from ultralytics)
  Using cached psutil-7.2.2-cp37-abi3-win_amd64.whl.metadata (22 kB)
Collecting polars>=0.20.0 (from ultralytics)
  Using cached polars-1.43.2-py3-none-any.whl.metadata (11 kB)
Collecting nvidia-ml-py>=12.0.0 (from ultralytics)
  Using cached nvidia_ml_py-13.610.43-py3-none-any.whl.metadata (9.7 kB)
Collecting ultralytics-thop>=2.1.6 (from ultralytics)
  Using cached ultralytics_thop-2.1.6-py3-none-any.whl.metadata (13 kB)
Collecting python-dateutil>=2.8.2 (from pandas)
  Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting pytz>=2020.1 (from pandas)
  Downloading pytz-2026.3.post1-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas)
  Using cached tzdata-2026.3-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Using cached contourpy-1.3.2-cp310-cp310-win_amd64.whl.metadata (5.5 kB)
Collecting cycler>=0.10 (from matplotlib)
  Using cached cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib)
  Using cached fonttools-4.63.0-cp310-cp310-win_amd64.whl.metadata (121 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Using cached kiwisolver-1.5.0-cp310-cp310-win_amd64.whl.metadata (5.2 kB)
Requirement already satisfied: packaging>=20.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from matplotlib) (26.2)
```

```
  Using cached pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Collecting polars-runtime-32==1.43.2 (from polars>=0.20.0->ultralytics)
  Using cached polars_runtime_32-1.43.2-cp310-abi3-win_amd64.whl.metadata (1.5 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas)
  Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting charset_normalizer<4,>=2 (from requests>=2.23.0->ultralytics)
  Using cached charset_normalizer-3.4.9-cp310-cp310-win_amd64.whl.metadata (42 kB)
Collecting idna<4,>=2.5 (from requests>=2.23.0->ultralytics)
  Using cached idna-3.18-py3-none-any.whl.metadata (6.1 kB)
Collecting urllib3<3,>=1.26 (from requests>=2.23.0->ultralytics)
  Using cached urllib3-2.7.0-py3-none-any.whl.metadata (6.9 kB)
Collecting certifi>=2023.5.7 (from requests>=2.23.0->ultralytics)
  Using cached certifi-2026.7.22-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: typing-extensions>=4.10.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=1.8.0->ultralytics) (4.16.0)
Requirement already satisfied: setuptools>=77.0.3 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=1.8.0->ultralytics) (83.0.0)
Requirement already satisfied: sympy>=1.13.3 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=1.8.0->ultralytics) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=1.8.0->ultralytics) (3.4.2)
Requirement already satisfied: jinja2 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=1.8.0->ultralytics) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=1.8.0->ultralytics) (2026.7.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics) (1.3
.0)
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from jinja2->torch>=1.8.0->ultralytics) (3.0.3)
Using cached ultralytics-8.4.115-py3-none-any.whl (1.4 MB)
Using cached pandas-2.3.3-cp310-cp310-win_amd64.whl (11.3 MB)
Using cached matplotlib-3.10.9-cp310-cp310-win_amd64.whl (8.2 MB)
Using cached contourpy-1.3.2-cp310-cp310-win_amd64.whl (221 kB)
Using cached cycler-0.12.1-py3-none-any.whl (8.3 kB)
Using cached fonttools-4.63.0-cp310-cp310-win_amd64.whl (1.6 MB)
Using cached kiwisolver-1.5.0-cp310-cp310-win_amd64.whl (73 kB)
Using cached nvidia_ml_py-13.610.43-py3-none-any.whl (53 kB)
Using cached opencv_python-5.0.0.93-cp37-abi3-win_amd64.whl (44.0 MB)
Using cached polars-1.43.2-py3-none-any.whl (847 kB)
Using cached polars_runtime_32-1.43.2-cp310-abi3-win_amd64.whl (52.6 MB)
Using cached psutil-7.2.2-cp37-abi3-win_amd64.whl (137 kB)
Using cached pyparsing-3.3.2-py3-none-any.whl (122 kB)
Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading pytz-2026.3.post1-py2.py3-none-any.whl (508 kB)
Using cached pyyaml-6.0.3-cp310-cp310-win_amd64.whl (158 kB)
Using cached requests-2.34.2-py3-none-any.whl (73 kB)
Using cached charset_normalizer-3.4.9-cp310-cp310-win_amd64.whl (162 kB)
Using cached idna-3.18-py3-none-any.whl (65 kB)
Using cached urllib3-2.7.0-py3-none-any.whl (131 kB)
Using cached certifi-2026.7.22-py3-none-any.whl (136 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
```

```
Installing collected packages: pytz, nvidia-ml-py, urllib3, tzdata, six, pyyaml, pyparsing, psutil, polars-runtime-32, opencv-python, kiwisolver, idna, fonttools, cycle
r, contourpy, charset_normalizer, certifi, requests, python-dateutil, polars, ultralytics-thop, pandas, matplotlib, ultralytics
Successfully installed certifi-2026.7.22 charset_normalizer-3.4.9 contourpy-1.3.2 cycler-0.12.1 fonttools-4.63.0 idna-3.18 kiwisolver-1.5.0 matplotlib-3.10.9 nvidia-ml-
py-13.610.43 opencv-python-5.0.0.93 pandas-2.3.3 polars-1.43.2 polars-runtime-32-1.43.2 psutil-7.2.2 pyparsing-3.3.2 python-dateutil-2.9.0.post0 pytz-2026.3.post1 pyyam
l-6.0.3 requests-2.34.2 six-1.17.0 tzdata-2026.3 ultralytics-8.4.115 ultralytics-thop-2.1.6 urllib3-2.7.0
```

```
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>pip install ultralytics pycr
yptodome pillow numpy pandas matplotlib scipy
Requirement already satisfied: ultralytics in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (8.4.115)
Requirement already satisfied: pycryptodome in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (3.23.0)
Requirement already satisfied: pillow in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (12.3.0)
Requirement already satisfied: numpy in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (2.2.6)
Requirement already satisfied: pandas in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (2.3.3)
Requirement already satisfied: matplotlib in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (3.10.9)
Requirement already satisfied: scipy in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (1.15.3)
Requirement already satisfied: filelock>=3.16.1 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (3.32.2)
Requirement already satisfied: opencv-python!=4.13.0.90,>=4.6.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (5.0.0.93)
Requirement already satisfied: pyyaml>=5.3.1 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (6.0.3)
Requirement already satisfied: requests>=2.23.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (2.34.2)
Requirement already satisfied: torch>=1.8.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (2.13.0)
Requirement already satisfied: torchvision>=0.9.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (0.28.0)
Requirement already satisfied: psutil>=5.8.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (7.2.2)
Requirement already satisfied: polars>=0.20.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (1.43.2)
Requirement already satisfied: nvidia-ml-py>=12.0.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (13.610.43)
Requirement already satisfied: ultralytics-thop>=2.1.6 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from ultralytics) (2.1.6)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from pandas) (2026.3.post1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from pandas) (2026.3)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from matplotlib) (1.3.2)
Requirement already satisfied: cycler>=0.10 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from matplotlib) (26.2)
Requirement already satisfied: pyparsing>=3 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: polars-runtime-32==1.43.2 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from polars>=0.20.0->ultralytics) (1.43.2)
Requirement already satisfied: six>=1.5 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from requests>=2.23.0->ultralytics) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from requests>=2.23.0->ultralytics) (3.18)
Requirement already satisfied: urllib3<3,>=1.26 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from requests>=2.23.0->ultralytics) (2.7.0)
Requirement already satisfied: certifi>=2023.5.7 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from requests>=2.23.0->ultralytics) (2026.7.22)
Requirement already satisfied: typing-extensions>=4.10.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=1.8.0->ultralytics) (4.16.0)
Requirement already satisfied: setuptools>=77.0.3 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=1.8.0->ultralytics) (83.0.0)
Requirement already satisfied: sympy>=1.13.3 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=1.8.0->ultralytics) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=1.8.0->ultralytics) (3.4.2)
Requirement already satisfied: jinja2 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=1.8.0->ultralytics) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=1.8.0->ultralytics) (2026.7.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics) (1.3
.0)
```

```
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from jinja2->torch>=1.8.0->ultralytics) (3.0.3)
```

- Also I installed **torchmetrics [image]**

```
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>pip install torchmetrics[ima
ge]
Collecting torchmetrics[image]
  Downloading torchmetrics-1.9.0-py3-none-any.whl.metadata (23 kB)
Requirement already satisfied: numpy>1.20.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torchmetrics[image]) (2.2.6)
Requirement already satisfied: packaging>17.1 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torchmetrics[image]) (26.2)
Requirement already satisfied: torch>=2.0.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torchmetrics[image]) (2.13.0)
Collecting lightning-utilities>=0.15.3 (from torchmetrics[image])
  Downloading lightning_utilities-0.15.3-py3-none-any.whl.metadata (5.5 kB)
Collecting torch-fidelity<=0.4.0 (from torchmetrics[image])
  Downloading torch_fidelity-0.4.0-py3-none-any.whl.metadata (2.1 kB)
Requirement already satisfied: torchvision>=0.15.1 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torchmetrics[image]) (0.28.0)
Requirement already satisfied: scipy>1.0.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torchmetrics[image]) (1.15.3)
Requirement already satisfied: pillow in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch-fidelity<=0.4.0->torchmetrics[image]) (12.3.0)
Collecting tqdm (from torch-fidelity<=0.4.0->torchmetrics[image])
  Using cached tqdm-4.70.0-py3-none-any.whl.metadata (57 kB)
Requirement already satisfied: typing_extensions in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from lightning-utilities>=0.15.3->torchmetrics[image
]) (4.16.0)
Requirement already satisfied: filelock in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=2.0.0->torchmetrics[image]) (3.32.2)
Requirement already satisfied: setuptools>=77.0.3 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=2.0.0->torchmetrics[image]) (83.0.0)
Requirement already satisfied: sympy>=1.13.3 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=2.0.0->torchmetrics[image]) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=2.0.0->torchmetrics[image]) (3.4.2)
Requirement already satisfied: jinja2 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=2.0.0->torchmetrics[image]) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from torch>=2.0.0->torchmetrics[image]) (2026.7.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from sympy>=1.13.3->torch>=2.0.0->torchmetrics[imag
e]) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (from jinja2->torch>=2.0.0->torchmetrics[image]) (3.0.3
)
Collecting colorama (from tqdm->torch-fidelity<=0.4.0->torchmetrics[image])
  Using cached colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Downloading torchmetrics-1.9.0-py3-none-any.whl (983 kB)
   ---------------------------------------- 983.4/983.4 kB 1.5 MB/s  0:00:00
Downloading torch_fidelity-0.4.0-py3-none-any.whl (85 kB)
Downloading lightning_utilities-0.15.3-py3-none-any.whl (31 kB)
Using cached tqdm-4.70.0-py3-none-any.whl (80 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Installing collected packages: lightning-utilities, colorama, tqdm, torchmetrics, torch-fidelity
Successfully installed colorama-0.4.6 lightning-utilities-0.15.3 torch-fidelity-0.4.0 torchmetrics-1.9.0 tqdm-4.70.0
```

- Ultralytics provides **YOLOV8** and torchmetrics [image] provides **FID**.

- The verification output is like this.

```
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>python -c "from ultralytics
import YOLO; print(); print('===ULTRALYTICS_CHECK==='); print('ultralytics ok')"

===ULTRALYTICS_CHECK===
ultralytics ok

(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>python
Python 3.10.20 | packaged by conda-forge | (main, Jun 11 2026, 03:28:51) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> from ultralytics import YOLO
>>> print("VERIFICATION: ultralytics ok")
VERIFICATION: ultralytics ok
>>> exit()
```

- I realized that my training process had to run in CPU mode, because CUDA was returning an output of 'False'.

```
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>python -c "import torch; pri
nt('CUDA:', torch.cuda.is_available())"
CUDA: False
```

- I decrypted the CARLA synthetic encrypted dataset into decrypted format by using **decrypt_all_images.py** python script.

```
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>python decrypt_all_images.py

  Step 2 - Receive and Decrypt the CARLA Dataset
Source: F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder\output
Destination: F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder\decrypted_output

  Manifest loaded for town01 (8000 entries)
    Copied labels/ folder (2000 files, plain text - not encrypted)
    Decrypted & verified: semantic_f032208.png
town01: 8000 files decrypted & verified, 0 failed.

  Manifest loaded for town02 (8000 entries)
    Copied labels/ folder (2000 files, plain text - not encrypted)
    Decrypted & verified: semantic_f032016.png
town02: 8000 files decrypted & verified, 0 failed.

  Manifest loaded for town03 (8000 entries)
    Copied labels/ folder (2000 files, plain text - not encrypted)
    Decrypted & verified: semantic_f032096.png
town03: 8000 files decrypted & verified, 0 failed.

  Manifest loaded for town04 (8000 entries)
    Copied labels/ folder (2000 files, plain text - not encrypted)
    Decrypted & verified: semantic_f032048.png
town04: 8000 files decrypted & verified, 0 failed.

  Manifest loaded for town05 (8000 entries)
    Copied labels/ folder (2000 files, plain text - not encrypted)
    Decrypted & verified: semantic_f035120.png
town05: 8000 files decrypted & verified, 0 failed.

============================================================
  DONE. Total decrypted & verified: 40000
     Total failed: 0
     Output location: F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder\decrypted_output
```

- By using this python file I decrypted the encrypted files, and restored the original RGB images, Depth images and Semantic segmentation images.

- And also, by using **security_engine.py, rbac_gateway.py, ml_data_loader.py** python scripts I confirmed the encryption details (key, mode CBC, IV location , padding)

- Then checked the all class mask files in each of these 5 towns.

- Then get the verdict as “Every class_mask file across all 5 towns contains only valid class IDs (0 – 3)”.

```
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>python check_all_class_masks
.py
town01: checking 2000 class_mask files...
town02: checking 2000 class_mask files...
town03: checking 2000 class_mask files...
town04: checking 2000 class_mask files...
town05: checking 2000 class_mask files...

============================================================
 DONE. Total class_mask files checked: 10000

Files containing each class (out of 10000 total):
  0 (car):        7454 files (74.5%)
  1 (pedestrian): 2484 files (24.8%)
  2 (cyclist):    2198 files (22.0%)
  3 (background): 10000 files (100.0%)

Files with INVALID pixel values (outside 0-3): 0

VERDICT: PASS - every class_mask file across all 5 towns contains only valid class IDs (0-3).
```

2026/08/07

- I understood the characteristics of the dataset (**car:pedestrian:cyclist ratio , small – object problem**) before writing the training code by using **explore_data.py** python script.

```
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>python explore_data.py
frames 10000
frames with 0 objects 3264 (32.6%)
mean objects/frame 1.51

car           11008 median area    1059 px^2 below 32x32:  49.2%
pedestrian     2424 median area     128 px^2 below 32x32:  86.4%
cyclist        1652 median area     289 px^2 below 32x32:  67.3%
```

- Next, I did and completed my security component part. My security component is **Data Poisoning Detection.** In here, I checked the structural anomalies (**Invalid class IDs, out of range coordinates**) & statistical distribution outliers (**town to town class – rate deviations**) and after that I flagged and reported them.
- In here I send only clean data to model training part.
- For that, I installed **scipy numpy package** and used **poison_detection.py** & **prove_poison_detection.py** python scripts.

```
(base) C:\Users\Windows>conda activate utility_eval

(utility_eval) C:\Users\Windows>cd /d "F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder"

(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>python poison_detection.py
Per-town class distribution
  town01: class0=0.656 class1=0.314 class2=0.030
  town02: class0=0.634 class1=0.263 class2=0.103
  town03: class0=0.734 class1=0.147 class2=0.120
  town04: class0=0.763 class1=0.114 class2=0.123
  town05: class0=0.762 class1=0.119 class2=0.119

Outlier check (z-score across towns)

Total boxes scanned: 15084
Anomalies found: 0

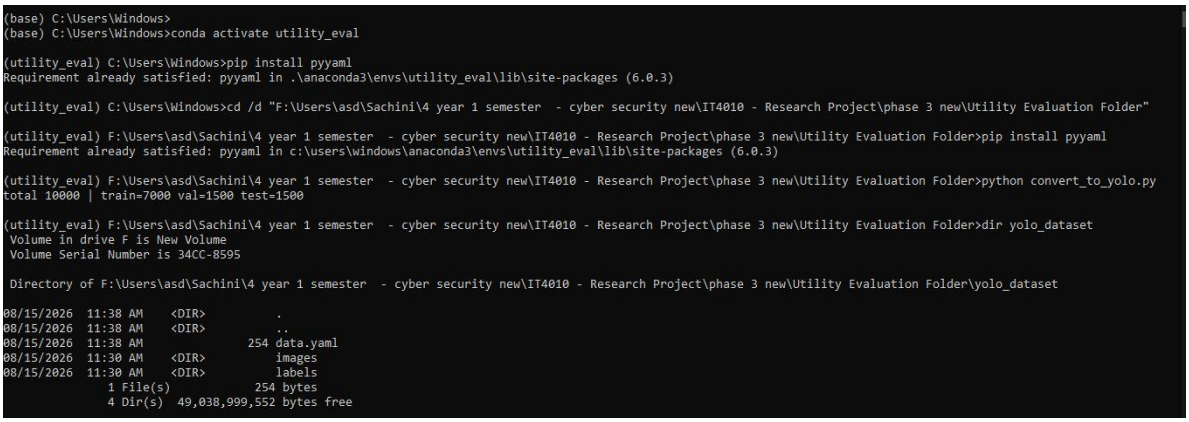
VERDICT: PASS — no evidence of tampering

(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>python prove_poison_detectio
n.py
Created deliberately poisoned file: test_poison\bbox_f000224.txt
Contents:
  1 1.5 0.5 0.2 0.2   <- x_center=1.5 is outside [0,1]
  7 0.5 0.5 0.1 0.1   <- class_id=7 does not exist (valid: 0,1,2)

Detector result: 3 anomalies found
  bbox_f000224.txt:1: x=1.5 outside [0,1]
  bbox_f000224.txt:1: box extends beyond frame horizontally
  bbox_f000224.txt:2: unknown class id 7

PROOF SUCCESSFUL: detector correctly flagged both injected defects.
```

2026/08/10	• Next, I took the decrypted RGB images and box labels, performed a 70/15/15 split for training, validation, and testing, and formatted them for YOLO training. For that I installed **pyyaml** and I used **convert_to_yolo.py** python script. • Then, I trained the YOLO model on synthetic data. For that I used **YOLOv8 (YOLO version 8) model** and used **train_yolo.py** script for that. The output seems like this.
2026/08/16	• Next, I measured the **Mean Average Precision (mAP) value**. This yielded three readable figures as output. Those are "**mAP50, mAP50 – 95, Per-class AP**". The output seems like this. • To receive the output like this I used **measure_map.py** & **size_breakdown_map.py** python scripts.



```
(base) C:\Users\Windows>
(base) C:\Users\Windows>conda activate utility_eval

(utility_eval) C:\Users\Windows>pip install pyyaml
Requirement already satisfied: pyyaml in .\anaconda3\envs\utility_eval\lib\site-packages (6.0.3)

(utility_eval) C:\Users\Windows>cd /d "F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder"

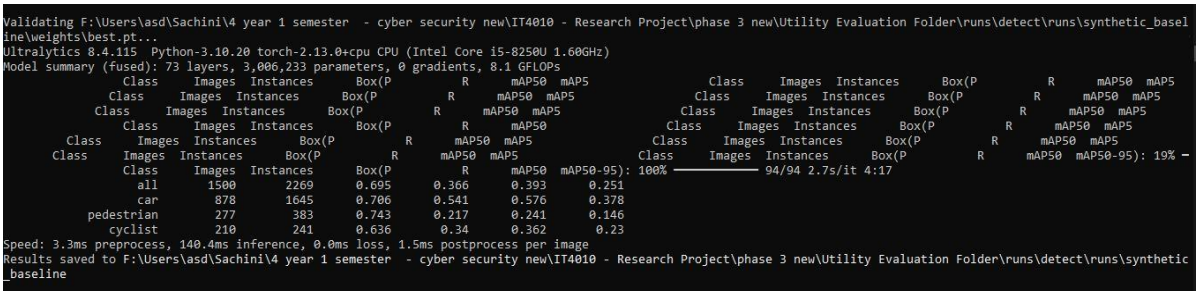
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>pip install pyyaml
Requirement already satisfied: pyyaml in c:\users\windows\anaconda3\envs\utility_eval\lib\site-packages (6.0.3)

(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>python convert_to_yolo.py
total 10000 | train=7000 val=1500 test=1500

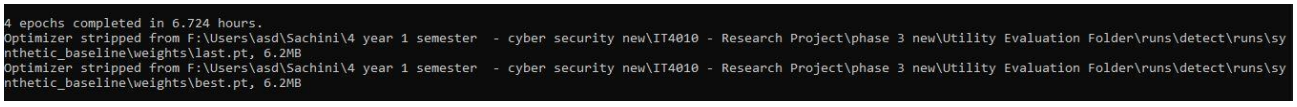
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>dir yolo_dataset
 Volume in drive F is New Volume
 Volume Serial Number is 34CC-8595

 Directory of F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder\yolo_dataset

08/15/2026  11:38 AM    <DIR>          .
08/15/2026  11:38 AM    <DIR>          ..
08/15/2026  11:38 AM               254 data.yaml
08/15/2026  11:30 AM    <DIR>          images
08/15/2026  11:30 AM    <DIR>          labels
               1 File(s)            254 bytes
               4 Dir(s)  49,038,999,552 bytes free
```



```
Validating F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder\runs\detect\runs\synthetic_basel
ine\weights\best.pt...
Ultralytics 8.4.115  Python-3.10.20 torch-2.13.0+cpu CPU (Intel Core i5-8250U 1.60GHz)
Model summary (fused): 73 layers, 3,006,233 parameters, 0 gradients, 8.1 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP5                 Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5               Class     Images  Instances      Box(P          R      mAP50  mAP5
             Class     Images  Instances      Box(P          R      mAP50  mAP5               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50                Class     Images  Instances      Box(P          R      mAP50  mAP5
         Class     Images  Instances      Box(P          R      mAP50  mAP5               Class     Images  Instances      Box(P          R      mAP50  mAP5
       Class     Images  Instances      Box(P          R      mAP50  mAP5                Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 19% -
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━ 94/94 2.7s/it 4:17
                   all       1500       2269      0.695      0.366      0.393      0.251
                   car        878       1645      0.706      0.541      0.576      0.378
            pedestrian        277        383      0.743      0.217      0.241      0.146
                cyclist        210        241      0.636       0.34      0.362       0.23
Speed: 3.3ms preprocess, 140.4ms inference, 0.0ms loss, 1.5ms postprocess per image
Results saved to F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder\runs\detect\runs\synthetic
_baseline
```



```
4 epochs completed in 6.724 hours.
Optimizer stripped from F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder\runs\detect\runs\sy
nthetic_baseline\weights\last.pt, 6.2MB
Optimizer stripped from F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder\runs\detect\runs\sy
nthetic_baseline\weights\best.pt, 6.2MB
```

```
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>python size_breakdown_map.py

  Size-Stratified mAP Analysis (Small / Medium / Large)
Running inference on 1500 test images (this may take a while on CPU)...
  ...300/1500 images processed
  ...600/1500 images processed
  ...900/1500 images processed
  ...1200/1500 images processed
  ...1500/1500 images processed

============================================================
RESULTS - Size-Stratified AP50 (single IoU=0.5 threshold)
============================================================

--- SMALL objects ---
  car          AP50=0.4708  (GT instances: 812, detections: 43505)
  pedestrian   AP50=0.1313  (GT instances: 367, detections: 10110)
  cyclist      AP50=0.2044  (GT instances: 177, detections: 2661)
  mAP50 (mean)=0.2688

--- MEDIUM objects ---
  car          AP50=0.6726  (GT instances: 489, detections: 19892)
  pedestrian   AP50=0.4760  (GT instances: 50, detections: 1545)
  cyclist      AP50=0.5900  (GT instances: 46, detections: 1681)
  mAP50 (mean)=0.5796

--- LARGE objects ---
  car          AP50=0.7971  (GT instances: 412, detections: 3176)
  pedestrian   AP50=0.0000  (GT instances: 5, detections: 7)
  cyclist      AP50=0.5078  (GT instances: 26, detections: 154)
  mAP50 (mean)=0.4350
```

```
                 all       1500       2367      0.646      0.338      0.366      0.227
                 car        876       1711      0.712      0.573      0.608      0.398
          pedestrian        288        407      0.619      0.155      0.179     0.0955
             cyclist        224        249      0.608      0.286       0.31      0.188
Speed: 3.0ms preprocess, 160.6ms inference, 0.0ms loss, 2.0ms postprocess per imageResults saved to F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010
- Research Project\phase 3 new\Utility Evaluation Folder\runs\detect\val
mAP50 0.3655
mAP50-95 0.2273
car          AP50=0.6076 AP50-95=0.3981
pedestrian   AP50=0.1793 AP50-95=0.0955
cyclist      AP50=0.3096 AP50-95=0.1882
```

Measure mAP_results.txt - Notepad

File Edit Format View Help

```
Command: measure_map.py
         size_breakdown_map.py

Test split results - official mAP measurement

Instances: 1,500 images, 2,367 objects (test split)

mAP50: 0.3655
mAP50-95: 0.2273 ← Primary metric

car:        AP50=0.6076 AP50-95=0.3981 (the highest)
pedestrian: AP50=0.1793 AP50-95=0.0955 (the lowest)
cyclist:    AP50=0.3096 AP50-95=0.1882 (the middle)
```

2026/08/17

- Next I downloaded **KITTI 2D object detection dataset** (left colour images plus training labels) from **cvibs.net / datasets / kitti**. Then I converted Kitti labels into YOLO format by using **kitti_to_yolo.py** python script. Then I ran four experiments that matter and got the results like this.

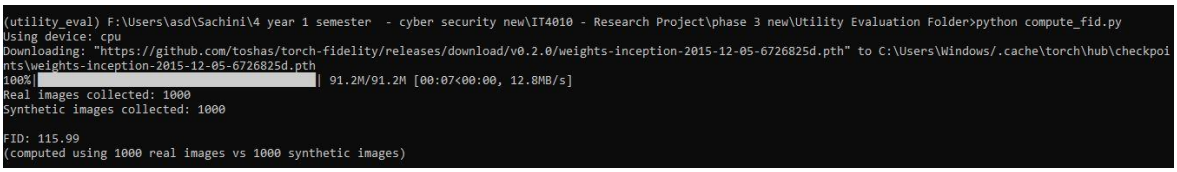
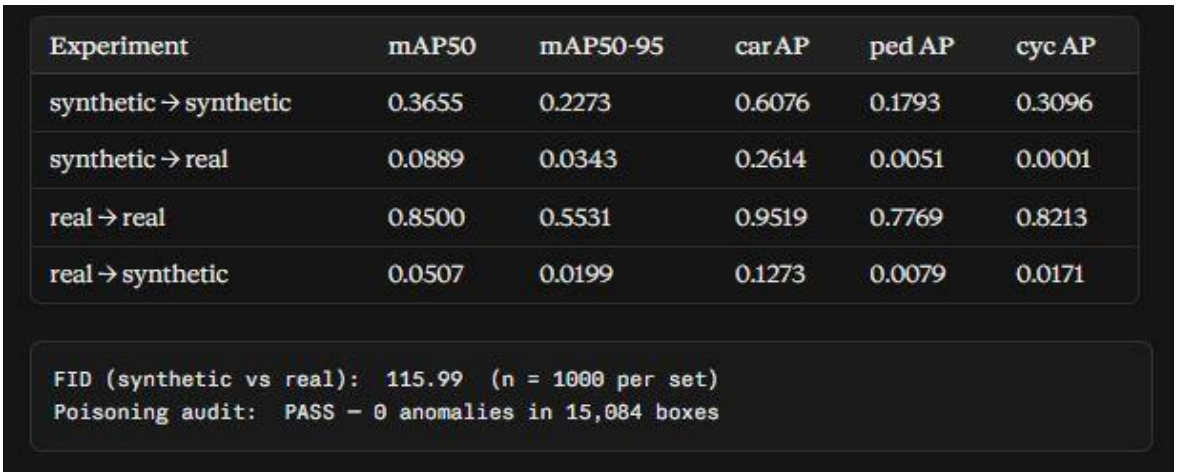
```
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010 - Research Project\phase 3 new\Utility Evaluation Folder>python kitti_to_yolo.py
Found 7481 image-label pairs
total 7481 | train=5236 val=1122 test=1123
```

```
============================================================
SUMMARY TABLE
============================================================
Experiment                  mAP50   mAP50-95    carAP    pedAP    cycAP
synthetic -> synthetic     0.3655     0.2273   0.6076   0.1793   0.3096
synthetic -> real          0.0889     0.0343   0.2614   0.0051   0.0001
real -> real               0.8500     0.5531   0.9519   0.7769   0.8213
real -> synthetic          0.0507     0.0199   0.1273   0.0079   0.0171
```

- The KITTI dataset's training completed output also like this.

```
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
               Class     Images  Instances      Box(P          R      mAP50  mAP5
0-95): 100% ━━━━━━━━━━━━ 71/71 1.1s/it 1:21
                 all       1122       6011        0.9      0.754      0.848
0.549
                 car       1029       5068      0.917      0.882      0.947
0.739
          pedestrian        281        679      0.879      0.602       0.74
 0.39
             cyclist        179        264      0.903      0.777      0.858
0.519
Speed: 1.4ms preprocess, 49.1ms inference, 0.0ms loss, 1.0ms postprocess per image
Results saved to F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT401
0 - Research Project\phase 3 new\Utility Evaluation Folder\runs\detect\runs\kitti_b
aseline

(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010
- Research Project\phase 3 new\Utility Evaluation Folder>
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010
- Research Project\phase 3 new\Utility Evaluation Folder>
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010
- Research Project\phase 3 new\Utility Evaluation Folder>
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010
- Research Project\phase 3 new\Utility Evaluation Folder>
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010
- Research Project\phase 3 new\Utility Evaluation Folder>
(utility_eval) F:\Users\asd\Sachini\4 year 1 semester  - cyber security new\IT4010
- Research Project\phase 3 new\Utility Evaluation Folder>
```

	• Then I calculated the "utility retained" value as a percentage. Then I got the output as 10.46% this is my core research finding. **utility retained % = (synthetic → real mAP50) / (real → real mAP50) × 100** **= 0.0889 / 0.8500 × 100** **= 10.46%**
2026/08/22	• Next, I calculated FID (Fréchet Inception Distance) value and got the output as **FID: 115.99** by using **compute_fid.py** python script. • FID measures how statistically similar two sets of images are. There are 3 ranges and each of these 3 ranges has their own interpretations. Those 3 ranges are like this: **<50 Reasonable, statistically similar** **50 – 200 Moderate difference** **>200 Very different**  • So, my output came under this "Moderate difference" interpretation.
2026/08/25	• So, the results table is like this: 

Experiment	mAP50	mAP50-95	car AP	ped AP	cyc AP
synthetic → synthetic	0.3655	0.2273	0.6076	0.1793	0.3096
synthetic → real	0.0889	0.0343	0.2614	0.0051	0.0001
real → real	0.8500	0.5531	0.9519	0.7769	0.8213
real → synthetic	0.0507	0.0199	0.1273	0.0079	0.0171

```
FID (synthetic vs real):  115.99  (n = 1000 per set)
Poisoning audit:  PASS — 0 anomalies in 15,084 boxes
```

Part 13: Results Table

Experiment	mAP50	mAP50-95	Car AP50	Ped AP50	Cyc AP50
synthetic -> synthetic	0.3655	0.2273	0.6076	0.1793	0.3096
synthetic -> real	0.0889	0.0343	0.2614	0.0051	0.0001
real -> real	0.8500	0.5531	0.9519	0.7769	0.8213
real -> synthetic	0.0507	0.0199	0.1273	0.0079	0.0171

FID (synthetic vs real): 115.99 (n = 1000 per set)

Poisoning audit: PASS, 0 anomalies in 15,084 boxes

Highlighted row = key sim-to-real result: 0.0889 / 0.8500 = 10.46% of real-data mAP50

- Supervisor's comments:
- Student progressing well in research project: (yes) / no
- If no:

- Project work to be carried out before next evaluation period:
- Signature: